Sahil Betal

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SUMMARY

Backend and Cloud Engineer skilled in designing scalable APIs, distributed systems, and high-performance backend services. Experienced in building production-grade platforms including honeypots, debugging engines, and analytics systems with measurable impact. Strong fundamentals in Python, FastAPI, Flask, microservices, asynchronous execution, and cloud-native engineering (AWS/GCP).

EDUCATION

Vishwakarma Institute of Technology *Aug 2023 – May 2027*
Bachelor of Technology in Computer Science *Pune, India*

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript

Backend & Systems: FastAPI, Flask, REST APIs, Node.js, Linux, Git, CI/CD, Nginx, Shell Scripting

Cloud & Tools: AWS, GCP (Vertex AI, Cloud Run), Docker, Kubernetes, MySQL, Pandas, NumPy, Plotly, Matplotlib, ETL Pipelines

SOFTWARE ENGINEERING PROJECTS

Custom Honeypot for Multi-Protocol Threat Analysis *2025*
Python, FastAPI, Networking, Systems Engineering

- Architected a multi-protocol honeypot (HTTP/SSH/SMB/FTP) capturing 150+ intrusion attempts from 30+ attacker IPs.
- Implemented event-driven pipelines for real-time threat classification, reaching 93% detection accuracy.
- Optimized concurrency, asynchronous I/O, and load handling to achieve uninterrupted operation during high-volume attacks.

CodeDebugger & Run *2025*
Python, FastAPI, AST, Systems Design

- Engineered an AST-based offline debugging engine enabling automated code analysis and error identification.
- Developed a secure, isolated execution backend supporting 70+ active users with controlled runtime environments.
- Accelerated debugging workflows by 40% through improved trace analysis, categorization, and reporting logic.

AURA — Data Visualization & Analytics Platform *2025*
Flask, Plotly, Backend Integration

- Designed a full-stack analytics system processing 1,000+ CSV datasets with automated data ingestion and visualization.
- Configured RESTful microservices powering real-time dashboards for 50+ simultaneous users.
- Boosted rendering and data pipeline throughput by 3× via caching, batching, and optimized query execution.

MediaPipe Face Landmarker *2024*
JavaScript, MediaPipe, Systems Integration

- Built a real-time facial landmark detector delivering 95%+ accuracy with sub-50ms inference latency.
- Implemented a dynamic rendering subsystem supporting 20+ avatars and 52 blend-shape animation controls.

ACHIEVEMENTS & PUBLICATIONS

Co-authored “Facial Landmark Detection and Avatar Animation” published in Springer (ICAIN 2025).

Published “Beta-Carotene: Molecular Structure C40H56” in IJSRR (ISSN: 2279-543X).

Ranked 13/110+ teams at InnoHack for backend architecture of CodeDebugger & Run.

CERTIFICATIONS

Cloud Technology Consultant — Coursera

AWS Cloud Solutions Architect — Coursera

NVIDIA Deep Learning Certificate — NVIDIA

Machine Learning Operations (MLOps) for Generative AI — Google

Machine Learning Operations (MLOps) with Vertex AI: Model Evaluation — Google